CIA 1

Part B

Question 1

Installing dependencies and importing libraries

In [1]:
```
!pip3 install pandas numpy scikit-learn openpyxl --break-system-packages
```

```
Requirement already satisfied: pandas in /Library/Frameworks/Python.framework/V
ersions/3.13/lib/python3.13/site-packages (2.3.1)
Requirement already satisfied: numpy in /Library/Frameworks/Python.framework/Ve
rsions/3.13/lib/python3.13/site-packages (2.3.1)
Requirement already satisfied: scikit-learn in /Library/Frameworks/Python.frame
work/Versions/3.13/lib/python3.13/site-packages (1.7.2)
Requirement already satisfied: openpyxl in /Library/Frameworks/Python.framewor
k/Versions/3.13/lib/python3.13/site-packages (3.1.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /Library/Frameworks/Py
thon.framework/Versions/3.13/lib/python3.13/site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Library/Frameworks/Python.frame
work/Versions/3.13/lib/python3.13/site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /Library/Frameworks/Python.fra
mework/Versions/3.13/lib/python3.13/site-packages (from pandas) (2025.2)
Requirement already satisfied: scipy>=1.8.0 in /Library/Frameworks/Python.frame
work/Versions/3.13/lib/python3.13/site-packages (from scikit-learn) (1.16.2)
Requirement already satisfied: joblib>=1.2.0 in /Library/Frameworks/Python.fram
ework/Versions/3.13/lib/python3.13/site-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /Library/Frameworks/Pyth
on.framework/Versions/3.13/lib/python3.13/site-packages (from scikit-learn)
(3.6.0)
Requirement already satisfied: et-xmlfile in /Library/Frameworks/Python.framewo
rk/Versions/3.13/lib/python3.13/site-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.5 in /Library/Frameworks/Python.framewor
k/Versions/3.13/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pand
as) (1.17.0)

[notice] A new release of pip is available: 25.1.1 -> 26.1.2
[notice] To update, run: pip3 install --upgrade pip
```

In [2]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
data = pd.read_excel("data/QP_Data_road_accident_dataset_with_missing.xlsx")
data.head()
```

Out[3]:

	Country	Year	Month	Day of Week	Time of Day	Urban/ Rural	Road Type	Weather Conditions	
0	USA	2002	October	Tuesday	Evening	Rural	Street	Windy	
1	UK	2014	December	Saturday	Evening	Urban	Street	Windy	16
2	USA	2012	July	Sunday	Afternoon	Urban	Highway	Snowy	34
3	UK	2017	May	Saturday	Evening	Urban	Main Road	Clear	48
4	Canada	2002	July	Tuesday	Afternoon	Rural	Highway	Rainy	34

5 rows × 30 columns

In [4]:
```
numeric = data.select_dtypes(include=np.number).keys()
print("Numeric data columns: ", numeric)
categorical = data.select_dtypes(include="object").keys()
print("Categorical data columns: ", categorical)
```

```
Numeric data columns:  Index(['Year', 'Visibility Level', 'Number of Vehicles I
nvolved',
       'Speed Limit', 'Driver Alcohol Level', 'Driver Fatigue',
       'Pedestrians Involved', 'Cyclists Involved', 'Number of Injuries',
       'Number of Fatalities', 'Emergency Response Time', 'Traffic Volume',
       'Insurance Claims', 'Medical Cost', 'Economic Loss',
       'Population Density'],
      dtype='object')
Categorical data columns:  Index(['Country', 'Month', 'Day of Week', 'Time of D
ay', 'Urban/Rural',
       'Road Type', 'Weather Conditions', 'Driver Age Group', 'Driver Gender',
       'Vehicle Condition', 'Accident Severity', 'Road Condition',
       'Accident Cause', 'Region'],
      dtype='object')
```

In [5]:
```
print("Data types of all columns: ", data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 132000 entries, 0 to 131999
Data columns (total 30 columns):
 #   Column                        Non-Null Count   Dtype  
---  ------                        --------------   -----  
 0   Country                       132000 non-null  object 
 1   Year                          132000 non-null  int64  
 2   Month                         132000 non-null  object 
 3   Day of Week                   132000 non-null  object 
 4   Time of Day                   132000 non-null  object 
 5   Urban/Rural                   132000 non-null  object 
 6   Road Type                     132000 non-null  object 
 7   Weather Conditions            132000 non-null  object 
 8   Visibility Level              125400 non-null  float64
 9   Number of Vehicles Involved   132000 non-null  int64  
 10  Speed Limit                   132000 non-null  int64  
 11  Driver Age Group              132000 non-null  object 
 12  Driver Gender                 132000 non-null  object 
 13  Driver Alcohol Level          104893 non-null  float64
 14  Driver Fatigue                132000 non-null  int64  
 15  Vehicle Condition             132000 non-null  object 
 16  Pedestrians Involved          132000 non-null  int64  
 17  Cyclists Involved             132000 non-null  int64  
 18  Accident Severity             132000 non-null  object 
 19  Number of Injuries            132000 non-null  int64  
 20  Number of Fatalities          132000 non-null  int64  
 21  Emergency Response Time       118824 non-null  float64
 22  Traffic Volume                125400 non-null  float64
 23  Road Condition                132000 non-null  object 
 24  Accident Cause                132000 non-null  object 
 25  Insurance Claims              132000 non-null  int64  
 26  Medical Cost                  106979 non-null  float64
 27  Economic Loss                 118800 non-null  float64
 28  Region                        132000 non-null  object 
 29  Population Density            125400 non-null  float64
dtypes: float64(7), int64(9), object(14)
memory usage: 30.2+ MB
Data types of all columns:  None
```

All the required information has been printed.

All the columns have appropriate data types.

Question 2a

Descriptive statistics for all the columns

In [6]:
```
data.describe()
```

Out[6]:

	Year	Visibility Level	Number of Vehicles Involved	Speed Limit	Driv Alcohol Lev
count	132000.000000	125400.000000	132000.000000	132000.000000	104893.0000
mean	2011.973348	275.104755	2.501227	74.544068	0.1104
std	7.198624	129.946180	1.117272	26.001448	0.0668
min	2000.000000	50.001928	1.000000	30.000000	0.0000
25%	2006.000000	162.422604	2.000000	52.000000	0.0538
50%	2012.000000	274.808977	3.000000	74.000000	0.1076
75%	2018.000000	388.070736	3.000000	97.000000	0.1614
max	2024.000000	499.999646	4.000000	119.000000	0.2499

The 'Economic Loss' column shows the highest variance from the mean with a value of 26642.146034 for it's standard deviation.

Question 2b

In [7]:
```
print("Medical Cost:")
print("Mean: ", data['Medical Cost'].mean())
print("Median: ", data['Medical Cost'].median())
```

```
Medical Cost:
Mean:  22999.326534801297
Median:  22687.63412
```

The median is lesser than the mean. Hence the distribution of 'Medical Cost' is skewed to the right or positively skewed.

Question 3a

In [8]:
```
data.isnull().sum()
```

```
Out[8]: Country                           0
        Year                              0
        Month                             0
        Day of Week                       0
        Time of Day                       0
        Urban/Rural                       0
        Road Type                         0
        Weather Conditions                0
        Visibility Level               6600
        Number of Vehicles Involved       0
        Speed Limit                       0
        Driver Age Group                  0
        Driver Gender                     0
        Driver Alcohol Level          27107
        Driver Fatigue                    0
        Vehicle Condition                 0
        Pedestrians Involved              0
        Cyclists Involved                 0
        Accident Severity                 0
        Number of Injuries                0
        Number of Fatalities              0
        Emergency Response Time       13176
        Traffic Volume                 6600
        Road Condition                    0
        Accident Cause                    0
        Insurance Claims                  0
        Medical Cost                  25021
        Economic Loss                 13200
        Region                            0
        Population Density             6600
        dtype: int64
```

In [9]:
```
print("Percentage of missing values: ")
data.isnull().sum() / data.shape[0] * 100.00
```

```
Percentage of missing values:
```

Out[9]:
```
Country                        0.000000
Year                           0.000000
Month                          0.000000
Day of Week                    0.000000
Time of Day                    0.000000
Urban/Rural                    0.000000
Road Type                      0.000000
Weather Conditions             0.000000
Visibility Level               5.000000
Number of Vehicles Involved    0.000000
Speed Limit                    0.000000
Driver Age Group               0.000000
Driver Gender                  0.000000
Driver Alcohol Level          20.535606
Driver Fatigue                 0.000000
Vehicle Condition              0.000000
Pedestrians Involved           0.000000
Cyclists Involved              0.000000
Accident Severity              0.000000
Number of Injuries             0.000000
Number of Fatalities           0.000000
Emergency Response Time        9.981818
Traffic Volume                 5.000000
Road Condition                 0.000000
Accident Cause                 0.000000
Insurance Claims               0.000000
Medical Cost                  18.955303
Economic Loss                 10.000000
Region                         0.000000
Population Density             5.000000
dtype: float64
```

Question 3b

Since 'Visibility Level', 'Population Density' and 'Traffic Volume' have 5 percent missing values, we can use mode imputation.

Question 3c

- Emergency Response Time (9.98%): Falls in the 5-10% range. We will use mean/median/mode substitution.
- Driver Alcohol Level (20.5%) & Medical Cost (18.9%): Have >10% missing values. We will use random sample imputation.

Question 3d

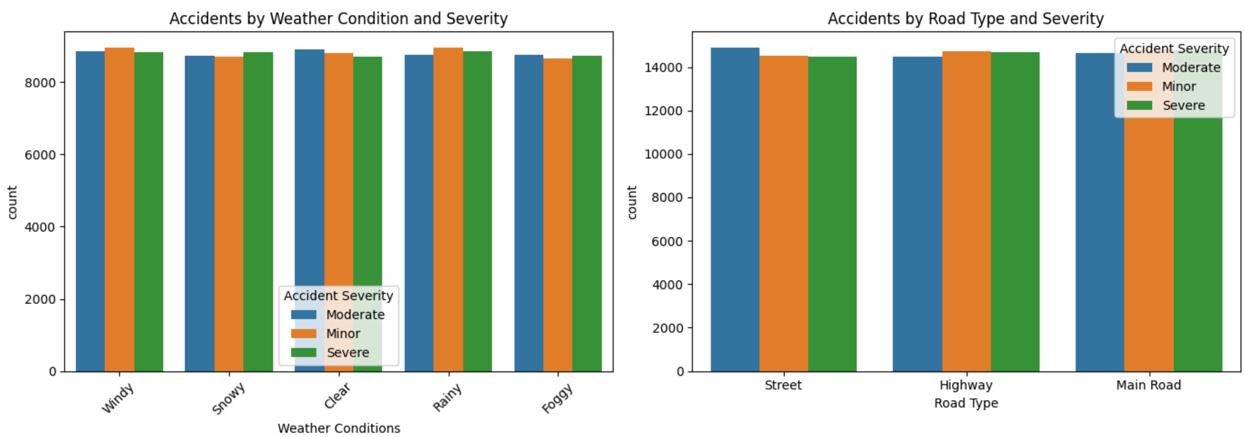
Missingness in 'Driver Alcohol Level' can be:

- MAR (Missing At Random): If the missingness depends on other observed variables like 'Time of Day' or 'Accident Severity'.

- MNAR (Missing Not At Random): If it depends on the alcohol level itself, like, highly intoxicated drivers refusing the test to avoid legal consequences.

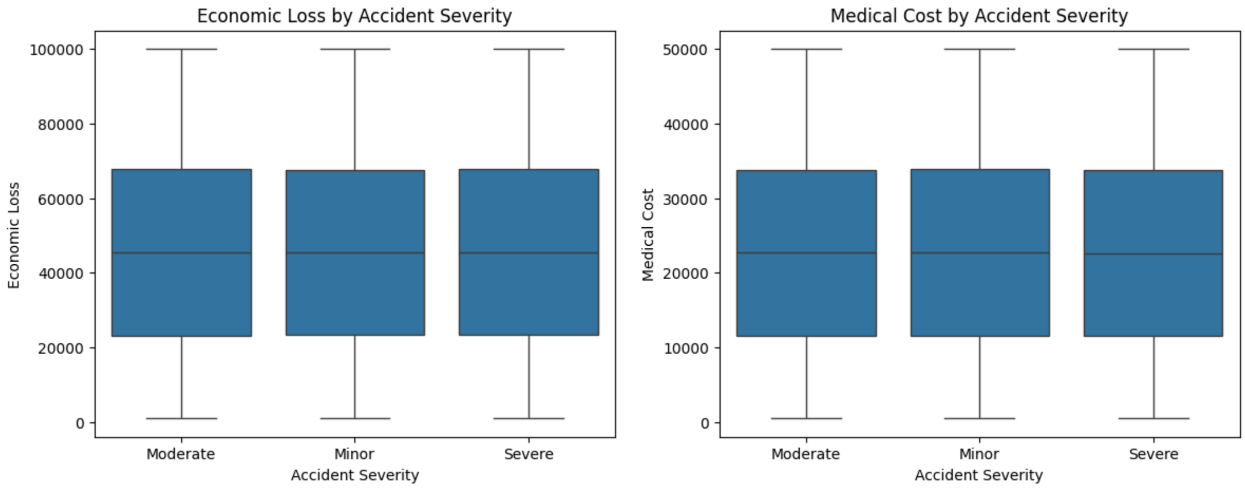
Question 4a

In [10]:
```
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.countplot(data=data, x='Weather Conditions', hue='Accident Severity', ax=a
axes[0].set_title('Accidents by Weather Condition and Severity')
axes[0].tick_params(axis='x', rotation=45)

sns.countplot(data=data, x='Road Type', hue='Accident Severity', ax=axes[1])
axes[1].set_title('Accidents by Road Type and Severity')
plt.tight_layout()
plt.show()
```



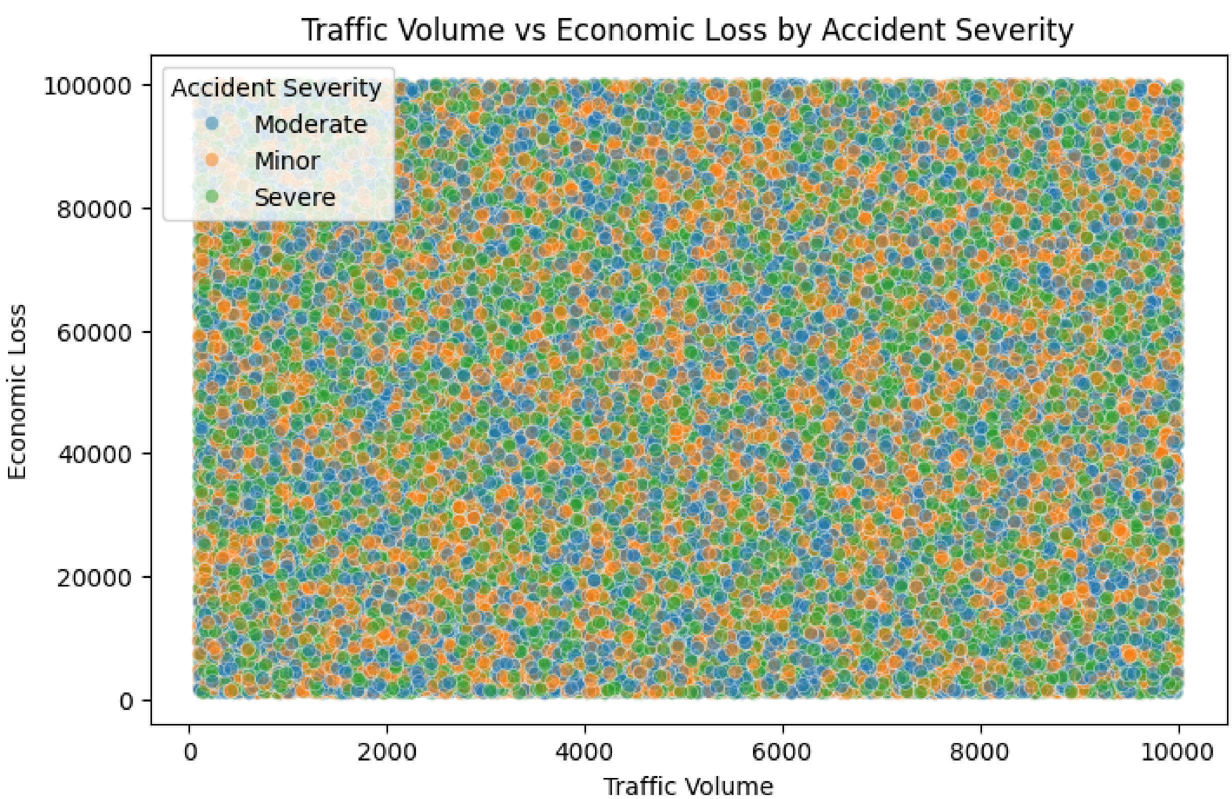
Question 4b

In [11]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.boxplot(data=data, x='Accident Severity', y='Economic Loss', ax=axes[0])
axes[0].set_title('Economic Loss by Accident Severity')

sns.boxplot(data=data, x='Accident Severity', y='Medical Cost', ax=axes[1])
axes[1].set_title('Medical Cost by Accident Severity')
plt.show()
```



Question 4c

In [12]:
```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=data, x='Traffic Volume', y='Economic Loss', hue='Acciden
plt.title('Traffic Volume vs Economic Loss by Accident Severity')
plt.show()
```



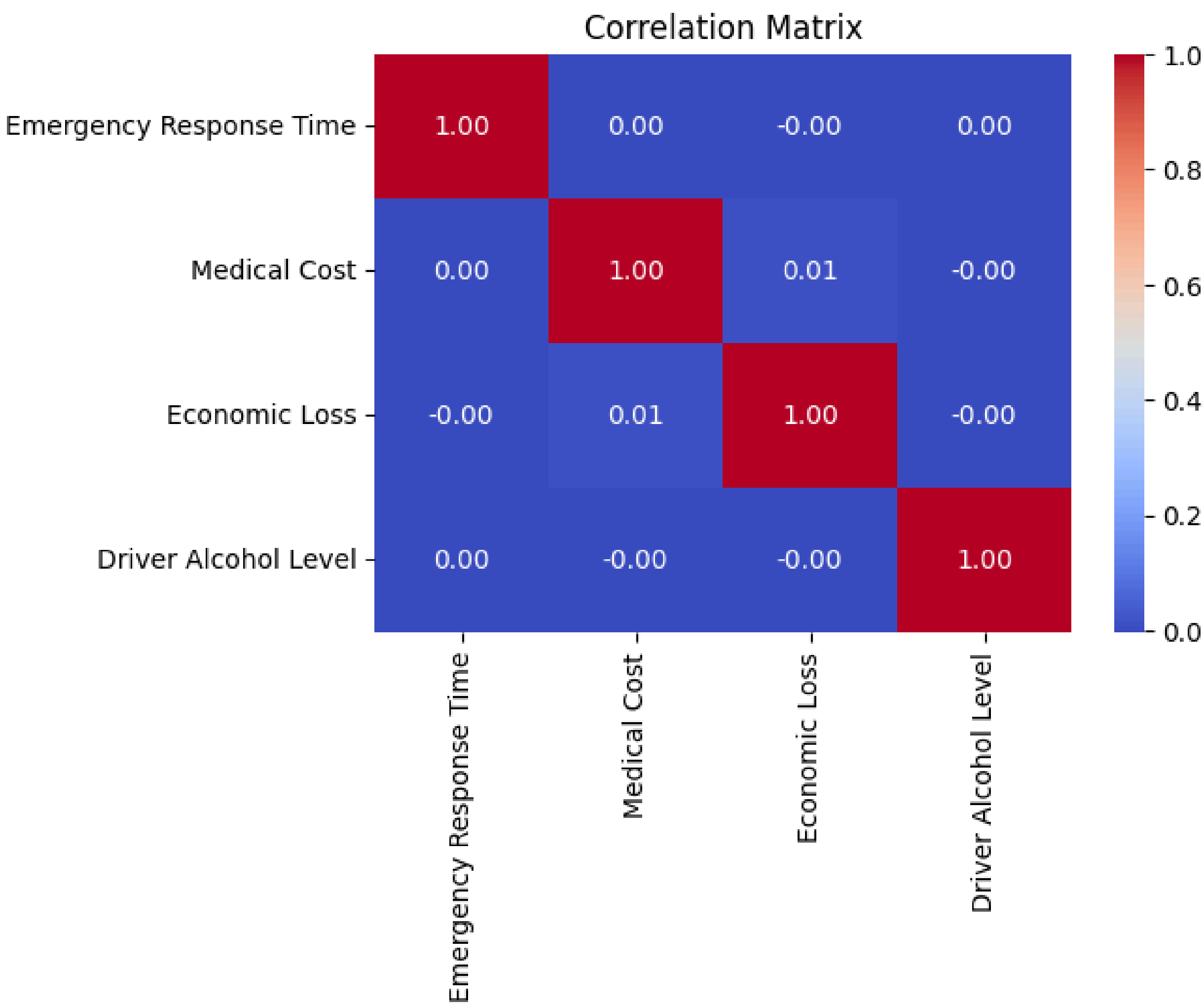
Question 4d

Intrpretations:

1. Weather and Severity: Severe accidents are more seen in foggy/snowy weather. Minor accidents can be seen as the major contributor.
2. Economic Loss by Severity: The boxplots show that Economic Loss and Medical Cost do not vary much across accident severities.
3. Traffic Volume vs Economic Loss: There is no clear linear relationship between traffic volume and economic loss.

Question 5

In [13]:
```
numeric_vars = ['Emergency Response Time', 'Medical Cost', 'Economic Loss', 'D
corr_matrix = data[numeric_vars].corr()

plt.figure(figsize=(6, 4))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```



Interpretation: The correlation matrix shows very weak or near-zero correlations among these variables, meaning they do not strongly linearly depend on each other.

Why categorical variables shouldn't be directly included: Categorical variables lack an inherent numerical order or magnitude. Mathematical operations used to calculate correlation (like variance and covariance) cannot be meaningfully applied to raw text labels without appropriate encoding.

Question 6a

In [14]:
```
cols_to_check = ['Medical Cost', 'Economic Loss', 'Traffic Volume', 'Emergency

for col in cols_to_check:
    Q1 = data[col].quantile(0.25)
    Q3 = data[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = data[(data[col] < lower_bound) | (data[col] > upper_bound)]
    print(f"{col} has {len(outliers)} outliers.")
```

```
Medical Cost has 0 outliers.
Economic Loss has 0 outliers.
Traffic Volume has 0 outliers.
Emergency Response Time has 0 outliers.
```

Question 6b

Examination of Medical Cost Outliers: The maximum value for Medical Cost is approximately 50,000, which can be due to extreme accident cases with fatal injuries. Therefore, these are likely meaningful extreme cases rather than data entry errors.

Question 7

a. Accident trends over time: Trends typically show fluctuations across years and may peak during certain months (e.g., winter months due to poor weather or holiday seasons with more travel).

b. Environmental and road conditions: Poor weather (snow/rain) and wet/icy roads increase the likelihood of accidents by reducing visibility and tire traction, acting as significant contributing factors.

c. Factors influencing severity and impact: Accident severity is often most significantly influenced by 'Speed Limit', 'Driver Alcohol Level', and 'Number of

Vehicles Involved', which directly translate to higher physical impact and subsequent medical/economic costs.

d. Geographic variation: Accident frequencies and impacts vary across regions due to differing population densities, infrastructure quality, and local traffic laws.